

TITLE: PSPJ - PROJECT

MODULE 2

ADAPTIVE QUIZ AND EXAM ENGINE

ABSTRACT

The Basic Adaptive Quiz is a simple Java program that changes the difficulty of questions based on the student's answers. The quiz starts at the Medium level and has Easy, Medium, and Hard questions. A correct answer increases the difficulty, while a wrong answer decreases it.

This project uses basic computational thinking concepts such as decomposition, pattern recognition, abstraction, and algorithm design. It also demonstrates Java concepts like variables, arrays, **if-else** statements, **for** loops, operators, and **Scanner** for user input. Overall, this project helped us understand how a real-world problem can be broken into simple steps and turned into a working Java program.

ADVANTAGES

Advantages of Using This Code

- It makes the quiz more interactive by changing the difficulty based on the user's answers.
- It helps students practice different levels of questions.
- It gives instant feedback when an answer is correct or wrong.
- It keeps track of the student's score automatically.
- It uses simple Java concepts, making it easy for beginners to understand.
- It can be easily modified by adding more questions or difficulty levels.
- It reduces the need for manual checking and scoring.

IMPLEMENTATION CODE:

```

import java.util.Scanner;
public class BasicAdaptiveQuiz {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int totalQuestions = 5;
        int score = 0;
        String difficulty = "Medium"; // starting level

        // Simple parallel arrays of questions per difficulty
        String[] easyQ = {"2 + 2 = ?", "5 - 3 = ?"};
        String[] mediumQ = {"6 x 7 = ?", "9 + 8 = ?"};
        String[] hardQ = {"12 x 12 = ?", "144 / 12 = ?"};

        int[] easyA = {4, 2};
        int[] mediumA = {42, 17};
        int[] hardA = {144, 12};

        int easyIndex = 0, mediumIndex = 0, hardIndex = 0;

        System.out.println("=== Basic Adaptive Quiz ===");

        // ---- ITERATION: fixed number of rounds ----
        for (int i = 1; i <= totalQuestions; i++) {

            String question;
            int correctAnswer;

            // ---- SELECTION: choose question based on current difficulty
            ----

            if (difficulty.equals("Easy")) {
                question = easyQ[easyIndex % easyQ.length];
                correctAnswer = easyA[easyIndex % easyA.length];
                easyIndex++;
            } else if (difficulty.equals("Medium")) {
                question = mediumQ[mediumIndex % mediumQ.length];
                correctAnswer = mediumA[mediumIndex % mediumA.length];
                mediumIndex++;
            }
        }
    }
}

```

```

        } else {
            question = hardQ[hardIndex % hardQ.length];
            correctAnswer = hardA[hardIndex % hardA.length];
            hardIndex++;
        }

        System.out.println("\nQ" + i + " [" + difficulty + "]: " +
question);

        System.out.print("Your answer: ");
        int userAnswer = sc.nextInt();

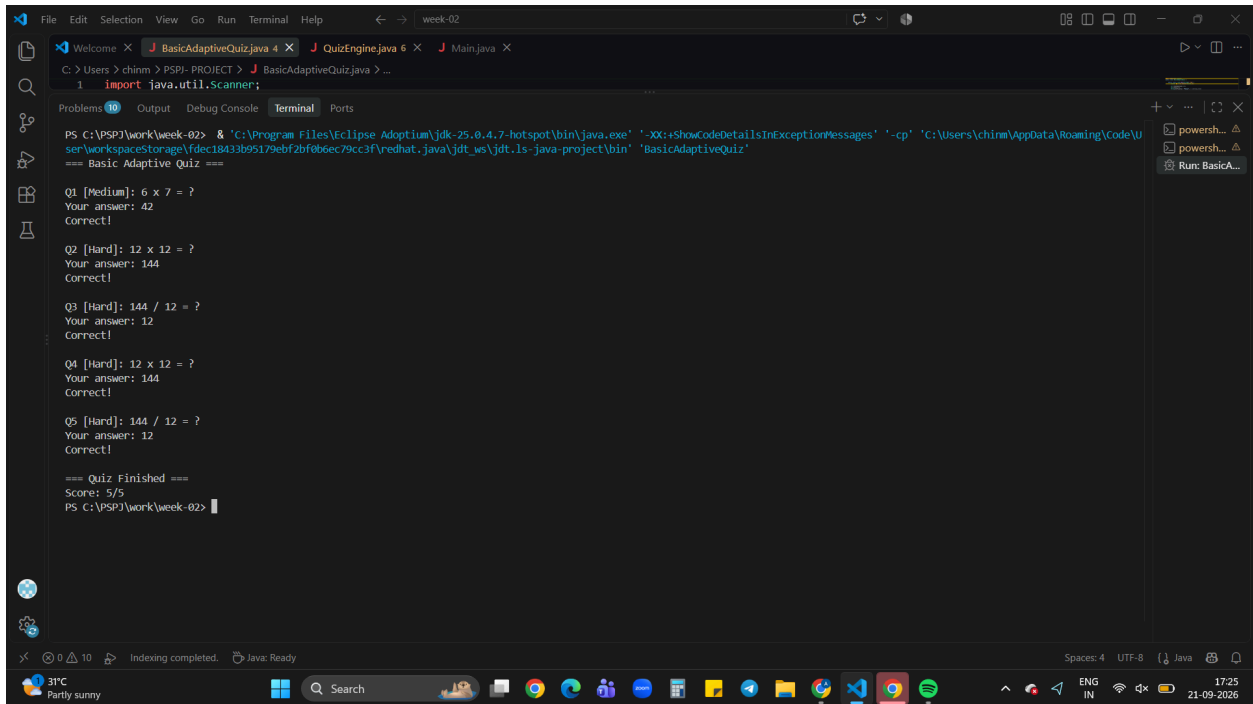
        // ---- SELECTION: check answer and adapt difficulty ----
        if (userAnswer == correctAnswer) {
            System.out.println("Correct!");
            score++;
            if (difficulty.equals("Easy")) {
                difficulty = "Medium";
            } else if (difficulty.equals("Medium")) {
                difficulty = "Hard";
            }
        } else {
            System.out.println("Wrong! Correct answer was " +
correctAnswer);

            if (difficulty.equals("Hard")) {
                difficulty = "Medium";
            } else if (difficulty.equals("Medium")) {
                difficulty = "Easy";
            }
        }
    }

    System.out.println("\n=== Quiz Finished ===");
    System.out.println("Score: " + score + "/" + totalQuestions);
    sc.close();
}
}

```

OUTPUT:



```
PS C:\PSP\work\week-02> & 'C:\Program Files\Eclipse Adoptium\jdk-25.0.4.7-hotspot\bin\java.exe' '-XX:ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\chinn\AppData\Roaming\Code\U
ser\workspaceStorage\fdcc18433b95179ebf2bf0b6ec79cc3f\redhat_java\jdt_ws\jdt.ls-java-project\bin' 'BasicAdaptiveQuiz'
=== Basic Adaptive Quiz ===

Q1 [Medium]: 6 x 7 = ?
Your answer: 42
Correct!

Q2 [Hard]: 12 x 12 = ?
Your answer: 144
Correct!

Q3 [Hard]: 144 / 12 = ?
Your answer: 12
Correct!

Q4 [Hard]: 12 x 12 = ?
Your answer: 144
Correct!

Q5 [Hard]: 144 / 12 = ?
Your answer: 12
Correct!

=== Quiz Finished ===
Score: 5/5
PS C:\PSP\work\week-02>
```

CONCLUSION:

The Basic Adaptive Quiz shows how Java can be used to create a simple and interactive quiz system. The code uses loops, conditions, arrays, and user input to change the difficulty based on the student's answers. This project helped us understand how computational thinking can be applied to solve a real-world problem in a simple way.